Deriving the Biomarker–Treatment Correlation Decay from First Principles

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Abstract

The `pmsimstats` data-generating process draws simulated trial data from a multivariate normal distribution whose correlation structure encodes a biomarker–treatment interaction. How the biomarker–biologic response (BM–BR) correlation should behave during off-drug periods with carryover has been a central methodological question, with three different implementations across the repository's history. This paper derives the correlation decay rule from first principles by tracing what happens to individual participant drug effects during pharmacokinetic clearance. The derivation shows that the correlation should decay at the same exponential rate as the mean drug effect, preserving the coefficient of variation of the biomarker-specific component. This result validates the core insight of the Ron Thomas proportional scaling while providing a principled parameterization via $\lambda_{cor} = \ln 2/t_{1/2}$.

Architecture scope. The λ_{cor} decay mechanism derived here is specific to Architecture B (MVN differential correlation), in which the biomarker-BR correlation decays exponentially during off-drug periods. Under Architecture A (direct mean moderation), the biomarker-treatment interaction is in the mean structure and does not involve correlation decay; the λ_{cor} parameter is unused. For a comparison of both DGP architectures, see `02-dgp-mean-moderation-vs-mvn.tex`. For implementation details of Architecture A, see `19-mean-moderation-implementation-notes.tex`.

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1 The Problem

The `pmsimstats` simulation generates participant outcomes by drawing from a multivariate normal (MVN) distribution. At each timepoint t, the biologic response (BR) factor has three properties specified in the MVN:

1. **Mean** $\mu_{\mathrm{BR},t}$: the expected drug effect, computed from the modified Gompertz function and carryover decay.
2. **Standard deviation** $\sigma_{\mathrm{BR}} = 8$: the between-person variability in drug response, constant across all timepoints.
3. **Correlation with biomarker** $\rho_t = \mathrm{Cor}(\mathrm{BM}, \mathrm{BR}_t)$: determines how much the biomarker predicts individual variation in drug response.

During on-drug periods, $\rho_t = c_{bm}$ (the specified biomarker correlation). During off-drug periods with carryover, the mean decays but the variance remains fixed. The question is: what should ρ_t be?

Three approaches have been implemented:

Step function (original, commit `42ac030`): $\rho_t = c_{bm}$ if $\mu_{\mathrm{BR},t} \neq 0$, else 0. Full correlation at any nonzero residual, regardless of magnitude.

Proportional scaling (Ron Thomas, commit `8609f12`): $\rho_t = (\mu_{\mathrm{BR},t}/\mu_{\mathrm{BR},t-1}) \cdot c_{bm}$. Correlation decays proportionally with the mean ratio between adjacent timepoints.

Independent exponential (current HEAD): $\rho_t = c_{bm} \cdot e^{-\lambda_{cor} \cdot t_{sd}}$. Correlation decays exponentially with time since discontinuation, with λ_{cor} as a free parameter.

This paper asks: is there a principled way to determine ρ_t from the structure of the problem, rather than treating it as an arbitrary modeling choice?

2 The Data-Generating Process

2.1 The MVN Framework

Each participant i receives a draw from the MVN. Focusing on the biomarker and BR factor at a single timepoint t:

$$\begin{pmatrix} \mathrm{BM}_i \\ \mathrm{BR}_{i,t} \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \mu_{\mathrm{BM}} \\ \mu_{\mathrm{BR},t} \end{pmatrix}, \begin{pmatrix} \sigma^2_{\mathrm{BM}} & \rho_t \sigma_{\mathrm{BM}} \sigma_{\mathrm{BR}} \\ \rho_t \sigma_{\mathrm{BM}} \sigma_{\mathrm{BR}} & \sigma^2_{\mathrm{BR}} \end{pmatrix} \right)$$

The conditional distribution of BR given a participant's biomarker value is:

$$\mathrm{BR}_{i,t} \mid \mathrm{BM}_i = bm \;\sim\; \mathcal{N}\left(\mu_{\mathrm{BR},t} + \rho_t \frac{\sigma_{\mathrm{BR}}}{\sigma_{\mathrm{BM}}} (bm - \mu_{\mathrm{BM}}),\; \sigma^2_{\mathrm{BR}}(1 - \rho_t^2) \right) \tag{1}$$

The conditional mean has two components:

- $\mu_{\mathrm{BR},t}$: the population-average drug effect (same for everyone)
- $\rho_t(\sigma_{\mathrm{BR}}/\sigma_{\mathrm{BM}})(bm - \mu_{\mathrm{BM}})$: the biomarker-specific deviation (varies by participant)

2.2 The Gompertz Mean Trajectory

The BR mean at on-drug timepoints follows the modified Gompertz:

$$\mu^{\text{on}}_{\text{BR},t} = f_G(\text{tod}_t;\ 10.99, 5, 0.42)$$

At off-drug timepoints, the raw Gompertz evaluates to zero (tod = 0), and carryover provides the mean:

$$\mu^{\text{off}}_{\text{BR},t} = \mu_{\text{BR},t-1} \cdot \left(\frac{1}{2}\right)^{t_{sd}/t_{1/2}} = \mu_{\text{BR},t-1} \cdot \alpha$$

where $\alpha = (1/2)^{t_{sd}/t_{1/2}}$ is the decay factor.

2.3 Parameter Values

Table 1: Key parameter values from the pilot RCT data

Parameter	Symbol	Value	Source
Biomarker mean	μ_{BM}	124.3 mmHg	Pilot systolic BP
Biomarker SD	σ_{BM}	15.4 mmHg	Pilot systolic BP
BR SD	σ_{BR}	8.0 CAPS points	Active−placebo variability
Biomarker correlation	c_{bm}	0–0.5	Simulation parameter

3 Derivation from Pharmacokinetic Principles

3.1 The Individual-Level Argument

Consider two participants during an on-drug period when the population BR mean is μ:

- Participant A has biomarker 1 SD above the population mean ($bm_A = \mu_{\text{BM}} + \sigma_{\text{BM}}$).
- Participant B has biomarker 1 SD below the population mean ($bm_B = \mu_{\text{BM}} - \sigma_{\text{BM}}$).

From Equation 1, their expected BR values are:

$$E[\text{BR}_A] = \mu + \rho \cdot \sigma_{\text{BR}} = \mu + c_{bm} \cdot 8 \tag{2}$$

$$E[\text{BR}_B] = \mu - \rho \cdot \sigma_{\text{BR}} = \mu - c_{bm} \cdot 8 \tag{3}$$

The biomarker-specific spread is:

$$\Delta_{\text{on}} = E[\text{BR}_A] - E[\text{BR}_B] = 2 \cdot c_{bm} \cdot \sigma_{\text{BR}} \tag{4}$$

3.2 What Happens at Drug Discontinuation?

When the drug is discontinued, the pharmacologic effect decays. If carryover is pharmacokinetic (drug molecules clear at a rate determined by the drug's half-life, independent of the patient's biomarker value), then *each participant's drug effect decays by the same fraction* α:

$$E[\mathrm{BR}_A^{\mathrm{off}}] = \alpha \cdot E[\mathrm{BR}_A] = \alpha \cdot (\mu + c_{bm} \cdot 8) \tag{5}$$

$$E[\mathrm{BR}_B^{\mathrm{off}}] = \alpha \cdot E[\mathrm{BR}_B] = \alpha \cdot (\mu - c_{bm} \cdot 8) \tag{6}$$

The population mean at the off-drug timepoint is:

$$\mu_{\mathrm{off}} = \alpha \cdot \mu$$

which matches the carryover formula in the DGP. The biomarker-specific spread is:

$$\Delta_{\mathrm{off}} = E[\mathrm{BR}_A^{\mathrm{off}}] - E[\mathrm{BR}_B^{\mathrm{off}}] = \alpha \cdot 2 \cdot c_{bm} \cdot \sigma_{\mathrm{BR}} = \alpha \cdot \Delta_{\mathrm{on}} \tag{7}$$

The spread decays by the same factor α as the mean.

3.3 Implied Correlation

From Equation 1, the biomarker-specific spread at the off-drug timepoint is also:

$$\Delta_{\mathrm{off}} = 2 \cdot \rho_{\mathrm{off}} \cdot \sigma_{\mathrm{BR}}$$

Setting this equal to Equation 7:

$$2 \cdot \rho_{\mathrm{off}} \cdot \sigma_{\mathrm{BR}} = \alpha \cdot 2 \cdot c_{bm} \cdot \sigma_{\mathrm{BR}}$$

$$\boxed{\rho_{\mathrm{off}} = \alpha \cdot c_{bm} = \left(\tfrac{1}{2}\right)^{t_{sd}/t_{1/2}} \cdot c_{bm}} \tag{8}$$

The BM–BR correlation at off-drug timepoints should decay by the same exponential factor as the BR mean. This is the central result.

3.4 Equivalence to Exponential Parameterization

The decay factor α can be written as:

$$\alpha = \left(\tfrac{1}{2}\right)^{t_{sd}/t_{1/2}} = e^{-(\ln 2/t_{1/2}) \cdot t_{sd}} = e^{-\lambda_{drug} \cdot t_{sd}}$$

where $\lambda_{drug} = \ln 2/t_{1/2}$ is the drug's elimination rate constant. Substituting into Equation 8:

$$\rho_{\mathrm{off}} = c_{bm} \cdot e^{-\lambda_{drug} \cdot t_{sd}}$$

This matches the current HEAD parameterization $\rho_t = c_{bm} \cdot e^{-\lambda_{cor} \cdot t_{sd}}$ when:

$$\boxed{\lambda_{cor} = \lambda_{drug} = \frac{\ln 2}{t_{1/2}}} \tag{9}$$

The correlation decay rate should equal the drug elimination rate constant.

4 Verification of Internal Consistency

4.1 Coefficient of Variation

The coefficient of variation (CV) of the biomarker-specific component measures how informative the biomarker is relative to the magnitude of the drug effect:

$$\text{CV} = \frac{\Delta}{\mu} = \frac{2 \cdot \rho \cdot \sigma_{\text{BR}}}{\mu}$$

On drug: $\text{CV}_{\text{on}} = 2 \cdot c_{bm} \cdot \sigma_{\text{BR}}/\mu$

Off drug with proportional scaling: $\text{CV}_{\text{off}} = 2 \cdot (\alpha \cdot c_{bm}) \cdot \sigma_{\text{BR}}/(\alpha \cdot \mu) = 2 \cdot c_{bm} \cdot \sigma_{\text{BR}}/\mu = \text{CV}_{\text{on}}$

The CV is preserved. The biomarker is equally informative *relative to the remaining effect*, which is the pharmacokinetically consistent expectation.

4.2 Numerical Example

N-of-1 Path A, $c_{bm} = 0.5$, $t_{1/2} = 1.0$ week:

Table 2: Conditional expectations under proportional correlation decay

	On drug (BD2, week 10)			Off drug (BD3, week 11)		
	Mean	$E[\text{BR}\| + 1\text{SD}]$	$E[\text{BR}\| - 1\text{SD}]$	Mean	$E[\text{BR}\| + 1\text{SD}]$	$E[\text{BR}\| - 1\text{SD}]$
μ_{BR}	10.19			5.09		
ρ	0.50			0.25		
$E[\text{BR}\|\text{BM}]$		14.19	6.19		7.09	3.09
Spread		8.0			4.0	
CV		0.785			0.786	

The spread halves (8.0 to 4.0) as the mean halves (10.19 to 5.09). The CV is preserved at 0.79. No participant has a negative expected BR. The conditional expectations are clinically plausible.

4.3 Comparison with Alternative Rules

Only the proportional scaling produces a spread that is consistent with the pharmacokinetic decay of individual effects. The step function ($\lambda_{cor} = 0$) eliminates the biomarker signal entirely. Full correlation ($\rho = c_{bm}$) maintains the on-drug spread despite the mean halving, implying the biomarker is twice as informative per unit of drug effect during carryover as during active treatment.

Table 3: Off-drug conditional expectations at BD3 ($\mu_{\text{BR}} = 5.09$, $c_{bm} = 0.5$) under three correlation rules

Rule	ρ	$E[\text{BR}\| + 1\text{SD}]$	$E[\text{BR}\| - 1\text{SD}]$	Issue
Step function ($\lambda_{cor} = 0$)	0	5.09	5.09	No biomarker signal
Proportional ($\lambda_{cor} = \lambda_{drug}$)	0.25	7.09	3.09	**Consistent**
Slow decay ($\lambda_{cor} = 1$)	0.18	6.57	3.61	Under-predicts spread
Full correlation ($\rho = c_{bm}$)	0.50	9.09	1.09	Over-predicts spread

5 Implications for the Publication's Findings

5.1 The Original Power Drop Was an Artifact

The original code used the step-function rule at short carryover half-lives ($t_{1/2} = 0.1, 0.2$ weeks). At these values, the carryover residual is negligible ($< 0.1\%$ of peak), but nonzero. The step function assigns full c_{bm} correlation to these timepoints, producing large biomarker-specific variation around a near-zero mean. This injects spurious noise into the interaction estimate.

Under proportional scaling, the correlation at these timepoints would be $\alpha \cdot c_{bm} \approx 0.001 \cdot c_{bm} \approx 0$, which is effectively the same as no correlation. The power drop disappears because the artifact disappears.

5.2 At Realistic Half-Lives, Power Should Drop Moderately

At clinically realistic half-lives ($t_{1/2} = 0.5$–1.0 weeks), the carryover residual is substantial (25–50% of peak) and the proportional correlation is meaningful ($\rho \approx 0.13$–$0.25 \cdot c_{bm}$). The off-drug timepoints carry partial biomarker interaction information that partially compensates for the carryover confounding.

The net effect on power depends on the balance between:

1. **Confounding (reduces power):** The carryover residual in the mean structure blurs the on-drug/off-drug contrast.
2. **Preserved signal (increases power):** The proportional correlation maintains biomarker-specific information during carryover periods.

A 200-replicate simulation with $\lambda_{cor} = 1$ (close to $\lambda_{drug} = 0.693$ at $t_{1/2} = 1.0$) showed flat power across carryover values, suggesting these two effects approximately cancel at the tested parameters.

5.3 The Correct λ_{cor} Value

From Equation 9, the correct value of λ_{cor} is not a fixed constant but depends on the carryover half-life being simulated:

A fixed λ_{cor} (such as the current default of 1.0) is only correct for $t_{1/2} \approx 0.693$ weeks (≈ 4.8 days). For other half-lives, it either over-decays the correlation (short $t_{1/2}$) or under-decays it (long $t_{1/2}$). The implementation should compute λ_{cor} from the carryover half-life rather than accepting it as a separate parameter.

Table 4: λ_{cor} values for different carryover half-lives

$t_{1/2}$ (weeks)	$\lambda_{cor} = \ln 2/t_{1/2}$
0.1	6.93
0.2	3.47
0.5	1.39
1.0	0.693
2.0	0.347

6 Relationship to the Three Historical Implementations

6.1 Step Function (Original)

The step function is the limiting case $\lambda_{cor} \to \infty$ for timepoints where $\mu = 0$ and $\lambda_{cor} = 0$ for timepoints where $\mu > 0$. It is a discontinuous approximation that fails when the mean is small but nonzero (the carryover regime). It produces clinically implausible conditional expectations (negative BR for low-biomarker participants) and artifact-driven power loss.

6.2 Ron Thomas Proportional Scaling

The proportional scaling computes $\rho_t = (\mu_t/\mu_{t-1}) \cdot c_{bm}$. Since $\mu_t/\mu_{t-1} = \alpha$ (the single-step decay factor), this is equivalent to $\rho_t = \alpha \cdot c_{bm}$, which is exactly Equation 8. The Ron Thomas implementation is the pharmacokinetically correct solution.

However, the implementation had three problems:

1. It used a ratio of adjacent means rather than the decay factor directly, requiring a $p > 1$ guard that inadvertently zeroed the correlation at timepoint 1.
2. It was bundled with a scale factor of 2 on the carryover decay that changed the meaning of $t_{1/2}$.
3. It was undocumented and unjustified in the commit.

6.3 Independent Exponential (Current HEAD)

The current parameterization $\rho_t = c_{bm} \cdot e^{-\lambda_{cor} \cdot t_{sd}}$ is the correct functional form. Setting $\lambda_{cor} = \ln 2/t_{1/2}$ recovers the pharmacokinetically derived result. The advantage over the Ron Thomas implementation is that it avoids the ratio formula, handles timepoint 1 correctly, and is explicit about the decay mechanism.

7 Recommended Implementation

The `generateData()` function should compute λ_{cor} automatically from the carryover half-life rather than accepting it as a separate parameter:

```
# In generateData():
if (modelparam$carryover_t1half > 0) {
  lambda_cor <- log(2) / modelparam$carryover_t1half
} else {
```

```
  lambda_cor <- 0
}
```

This ensures that the correlation decay is always consistent with the mean decay, regardless of the half-life being simulated. The separate λ_{cor} parameter can be retained as an override for sensitivity analysis (testing scenarios where the biomarker's predictive power decays faster or slower than the drug effect), but the default should be the pharmacokinetically derived value.

The BM–BR correlation assignment then becomes:

```
for (p in 1:nP) {
  n1 <- paste(trialdesign$timeptname[p], "br", sep = ".")
  if (d[p]$onDrug) {
    correlations[n1, 'bm'] <- modelparam$c.bm
    correlations['bm', n1] <- modelparam$c.bm
  } else if (d[p]$tsd > 0 && lambda_cor > 0) {
    decay <- exp(-lambda_cor * d$tsd[p])
    correlations[n1, 'bm'] <- modelparam$c.bm * decay
    correlations['bm', n1] <- modelparam$c.bm * decay
  }
}
```

8 The Remaining Variance Problem

The derivation in Section 3 assumes that the BR standard deviation $\sigma_{\mathrm{BR}} = 8$ is constant, which is the current implementation. This creates a situation where the conditional variance $\sigma^2_{\mathrm{BR}}(1 - \rho_t^2)$ is nearly unchanged during off-drug periods (since ρ_t is small), meaning participants still have large random fluctuations in BR around a decaying mean.

This is a deeper structural limitation of the MVN parameterization. The SD of 8 conflates two sources of variation:

1. **Trait variability:** Between-person differences in latent drug responsiveness, which exist before the drug is administered and persist regardless of drug status. This component should remain constant.
2. **State-dependent variability:** Variation in the acute drug effect, which should scale with the magnitude of the effect. This component should decay during carryover.

The current framework cannot separate these two sources. The proportional correlation scaling addresses the biomarker-specific component correctly (the spread decays with the mean), but the total variance remains inflated during off-drug periods. This means the conditional distribution has the correct center (proportional spread) but excessive width (fixed SD). The practical consequence is that the simulation adds more noise during off-drug periods than is clinically realistic, which may conservatively underestimate power.

Addressing this limitation would require a time-varying $\sigma_{\mathrm{BR},t}$ or a decomposition of the BR variance into trait and state components, either of which represents a substantial restructuring of the DGP. For the current application (comparing relative power across trial designs), the fixed-variance assumption is acceptable because it affects all designs similarly.

9 Conclusions

1. The BM–BR correlation during off-drug carryover periods should decay at the same exponential rate as the BR mean: $\rho_t = c_{bm} \cdot (1/2)^{t_{sd}/t_{1/2}}$. This follows from the pharmacokinetic principle that drug clearance affects all participants' effects by the same fraction, preserving the coefficient of variation of the biomarker-specific component.
2. This result is equivalent to setting $\lambda_{cor} = \ln 2/t_{1/2}$ in the current exponential parameterization, linking the correlation decay to the drug elimination rate constant.
3. The Ron Thomas proportional scaling (commit `8609f12`) implemented the correct principle but with an unnecessarily complex formula and three co-packaged changes (scale factor, timepoint-1 skip) that introduced new problems.
4. The recommended implementation computes λ_{cor} automatically from `carryover_t1half`, making the correlation decay consistent with the mean decay by construction, while retaining λ_{cor} as an optional override for sensitivity analysis.
5. The power drop reported in Hendrickson et al. (2020) at $t_{1/2} = 0.1$ and 0.2 weeks was an artifact of the step-function correlation rule, not a genuine consequence of carryover at those half-lives. At clinically realistic half-lives ($t_{1/2} = 0.5$–1.0 weeks), the proportional correlation decay partially compensates for carryover confounding, producing moderate rather than catastrophic power loss.
6. The fixed BR variance ($\sigma_{\mathrm{BR}} = 8$) is a known limitation that conservatively inflates noise during off-drug periods. It does not affect the validity of the correlation decay derivation but may underestimate power relative to a model with time-varying variance.

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